

Lecture 5: SQL JOINS

- Today: combining rows from two or more tables using a related column.
- An INNER JOIN returns only rows with matching values in both tables.
- Syntax: `SELECT * FROM A INNER JOIN B ON A.id = B.a_id.`
- Non-matching rows are excluded from an INNER JOIN entirely.

Outer JOINS

- A LEFT JOIN keeps every row from the left table, matched or not.
- A RIGHT JOIN keeps every row from the right table, matched or not.
- A FULL OUTER JOIN keeps unmatched rows from both sides, filling NULLs.
- Use LEFT JOIN whenever you must not lose rows from the primary table.

Self JOINS and JOIN performance

- A SELF JOIN joins a table to itself using two different aliases.
- Common use case: finding an employee's manager in the same table.
- JOIN performance depends heavily on indexes on the join columns.
- Always check EXPLAIN when a multi-table JOIN query feels slow.